MILITARY SPECIFICATION SHEET

POWDERS, PLASMA SPRAY
NICKEL ALUMINUM POWDER, TYPE I, COMPOSITION G, CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the powders described herein shall consist of this document and the latest issue of MIL-P-83348.

1. SCOPE

1.1 Scope. This specification sheet covers requirements intended for use in procurement of powders used to produce nickel aluminide coatings. These requirements are necessary to ensure the production of high quality nickel aluminide coatings.

2. APPLICABLE DOCUMENTS

This section is not applicable to this specification sheet.

3. REQUIREMENTS

3.1 Composition. Powder used to produce nickel aluminide coatings shall meet the composition requirements of MIL-P-83348 for a type I, class 2, composition G powder, except as specified by 3.1.1 and 3.1.2. Chemical composition determination and certified analysis shall be performed as specified in MIL-P-83348.

3.1.1 Particles. The powder particles shall consist of an aluminum core enclosed in a nickel jacket. Mechanical mixtures of nickel and aluminum or prealloyed nickel and aluminum powders are unacceptable.

3.1.2 Reaction during spraying. The powder shall react exothermically during deposition to form nickel aluminides.

3.2 Powder performance requirements. The powder shall be capable of producing acceptable coatings when deposited by either the thermospray or plasma spray process. Acceptable coatings are those which meet the property requirements specified in 3.2.1. Unacceptable coatings are those that either do not meet the requirements of 3.2.1 or possess defects, imperfections, or properties which render the coating unfit for use.

3.2.1 Coating properties. Powder when deposited either by the thermospray or plasma spray process must produce coatings having the properties specified in table I. The contractor shall furnish a certified analysis of the deposited coating showing compliance with the properties of table I.

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TABLE I. Coating property requirements.

Coating property	Thermo spray	Plasma spray
Coating structure	Dense sintered structure of nickel aluminides	
Melting point	No appreciable melting below 3000°F	
Bond strength	3000 psi	5000 psi
Porosity	5 percent or less	1.0 percent
Macrohardness	Rockwell B70 minimum	Rockwell B70 minimum
Coating density (approximately)	6.0 g per cm^3	6.9 g per cm^3

3.2.2 Deposition parameters. All deposition parameters used to produce coatings which meet the requirements of 3.2 and 3.2.1 must be furnished by the contractor. The parameters furnished must be capable of producing the required coating properties when used with METCO Inc. type 5P and 6P systems for thermospray and type 3M and 7M for plasma spray.

3.3 Rejections. Powder not conforming to the requirements specified herein will be subject to rejection. The Government reserves the right to perform any testing or inspection where such is deemed necessary to ensure conformance to the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions shall be as specified in MIL-P-83348.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking requirements shall be as specified in MIL-P-83348.

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6. NOTES

6.1 Comparable documents. Powder which meets the requirements specified herein will also meet the requirements of Pratt and Whitney Aircraft Corporation's specification PWA 1321 and General Electric Company's specification B50FF33.

Custodians:
Air Force - 20
Navy - SH
Army - MR

Preparing Activity:
Air Force - 20

Project MFFP-0310

Review activities:
Air Force - 99, 71
Navy - AS
Army - AR
Other - DLA-GS

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